

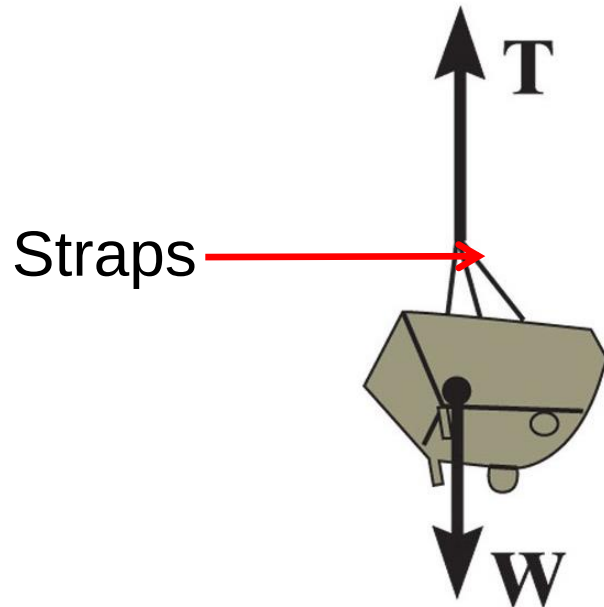
Topic 3: Equilibrium of a Particle



Applications

The crane is lifting a load at a steady rate

1. Will the straps be able to safely hold the load?
2. We need to specify the parameters of each cable to avoid breakage.



Contents

- Conditions for Equilibrium
- Free-Body Diagram
- Coplanar (2D) Force Systems
- Non-coplanar (3D) Force Systems

Equilibrium

- When the resultant of all of the forces applied to a particle is zero, the particle is in static equilibrium.

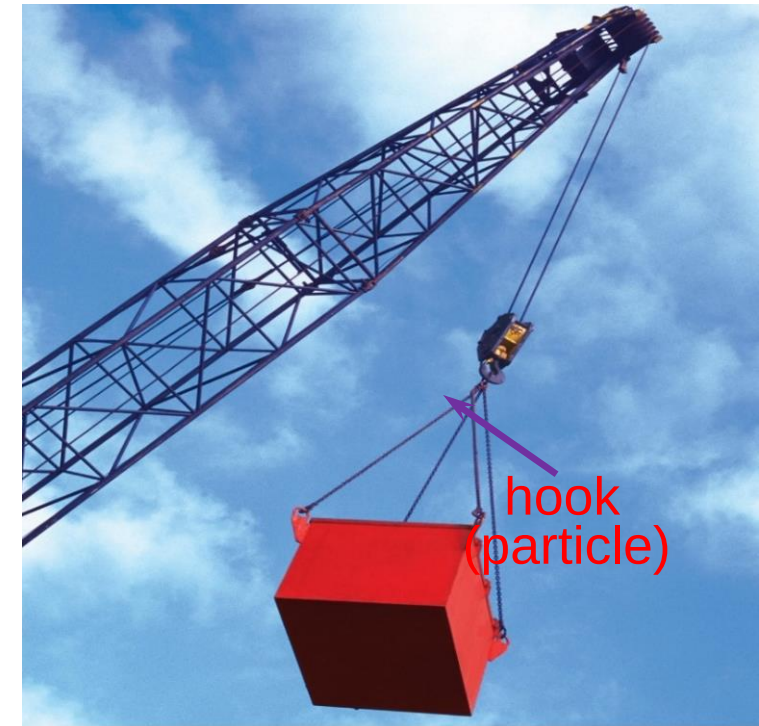
$$\sum \mathbf{F} = \mathbf{0}$$

OR

$$\mathbf{F}_R = \mathbf{0}$$

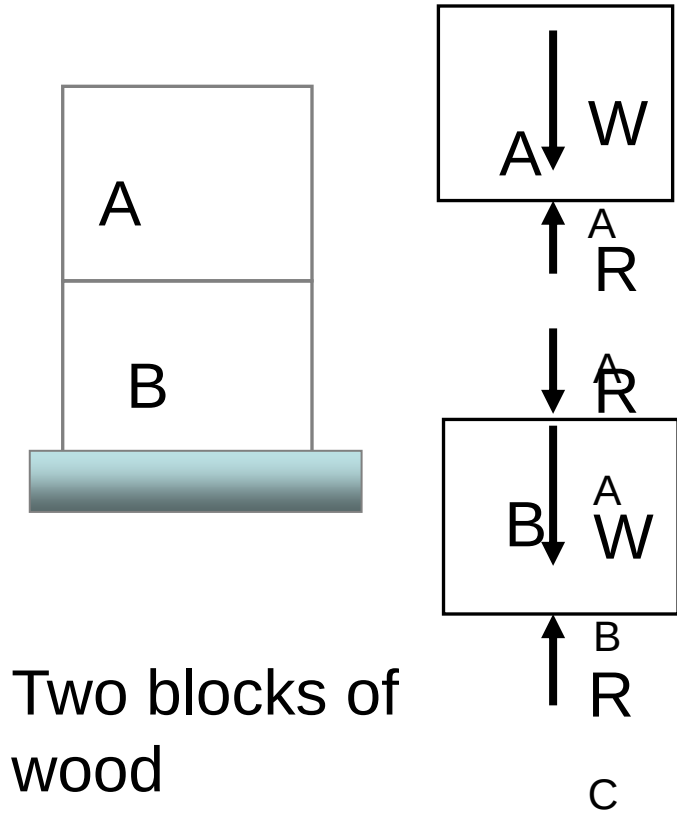
OR

$$(F_R)_x = \sum F_x = 0 \quad (F_R)_y = \sum F_y = 0 \quad (F_R)_z = \sum F_z = 0$$



Free-body Diagram

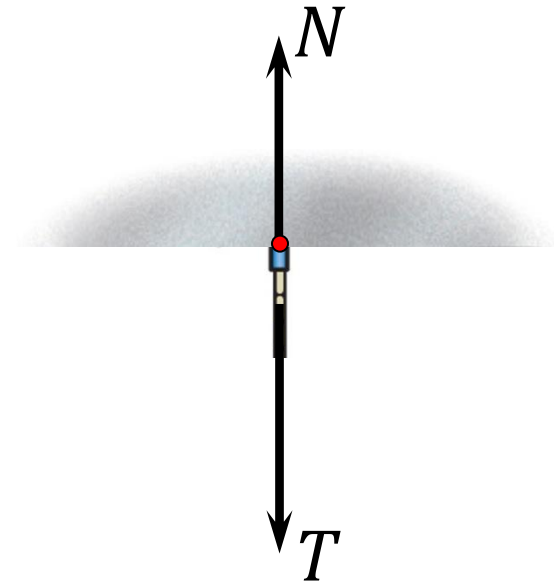
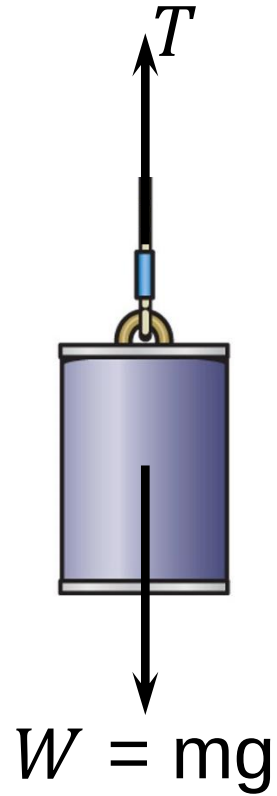
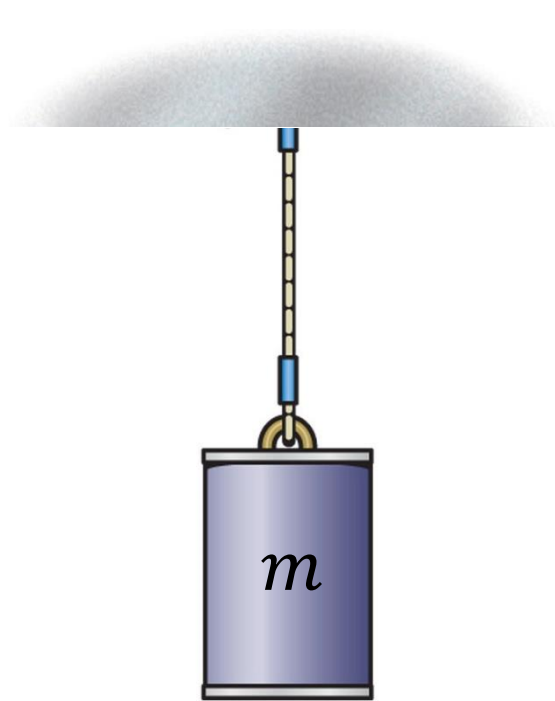
Steps for Drawing a Free-body Diagram



- Decide which body (or group of bodies) is under consideration, imagine it to be isolated from all other bodies, and sketch the outlined shape of the body.
- Indicate by means of arrows all external forces acting on the body. This should include (a) the weight of the body, (b) all external forces (c) reactions at supports and other contacts with other bodies.
- For each unknown force, indicate its point of application and assume a direction, if it is unknown.
- Include the dimensions and angles needed for resolving forces.
- The weight of a body always acts vertically downward through the centre of gravity of the body.

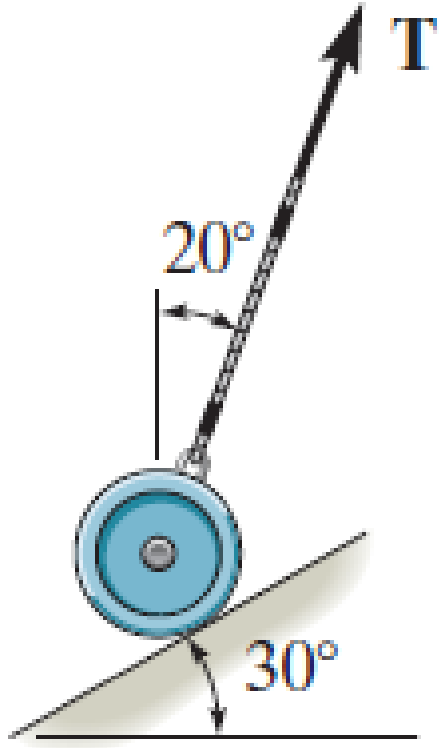
Free-body Diagram (Further Examples)

- Object (Particle) Supported by a Cable/Rope

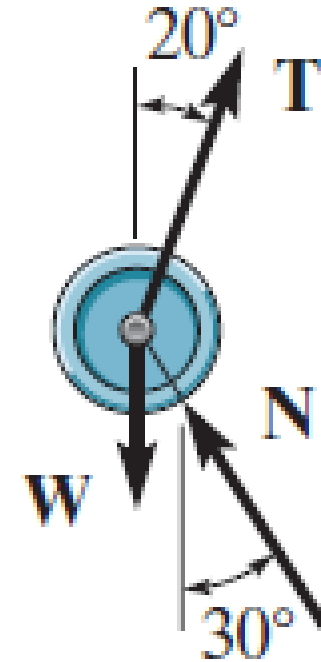


Free-body Diagram (Further Examples)

- Object (Particle) Resting on a Smooth Inclined Surface



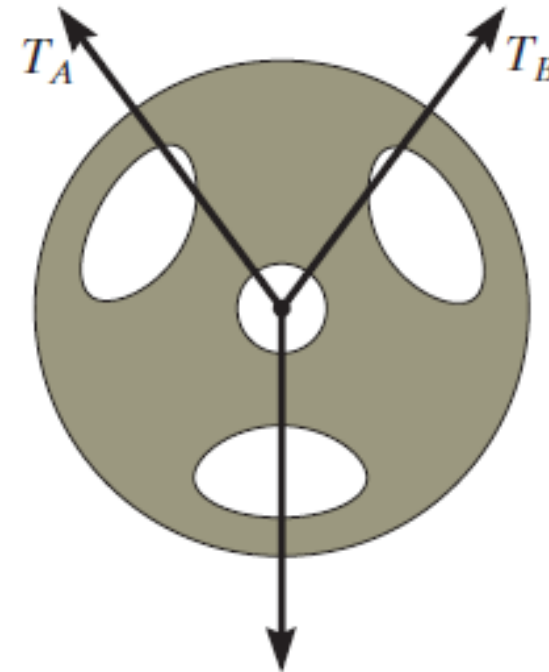
Space Diagram: A sketch showing the physical conditions of the problem.



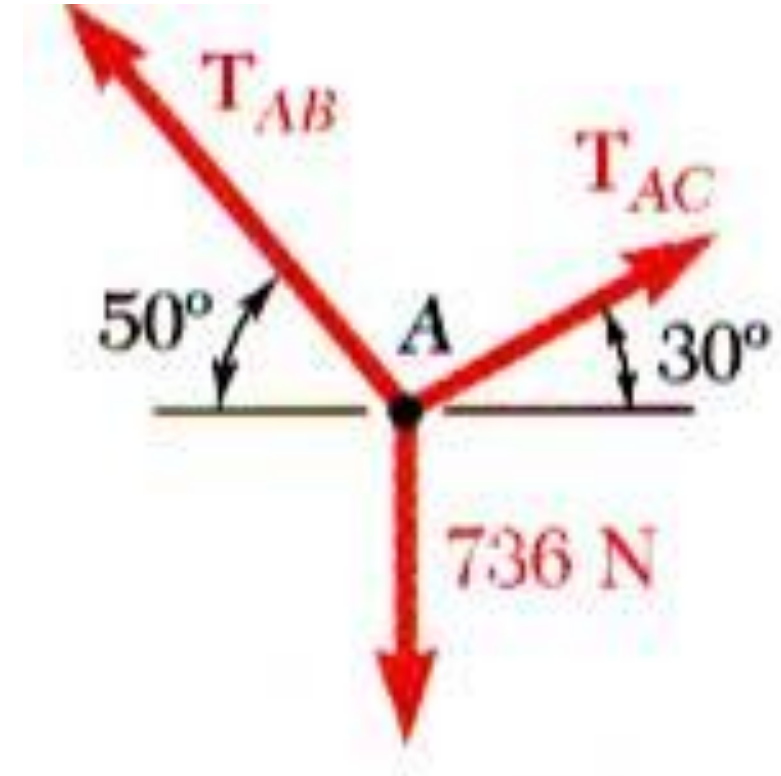
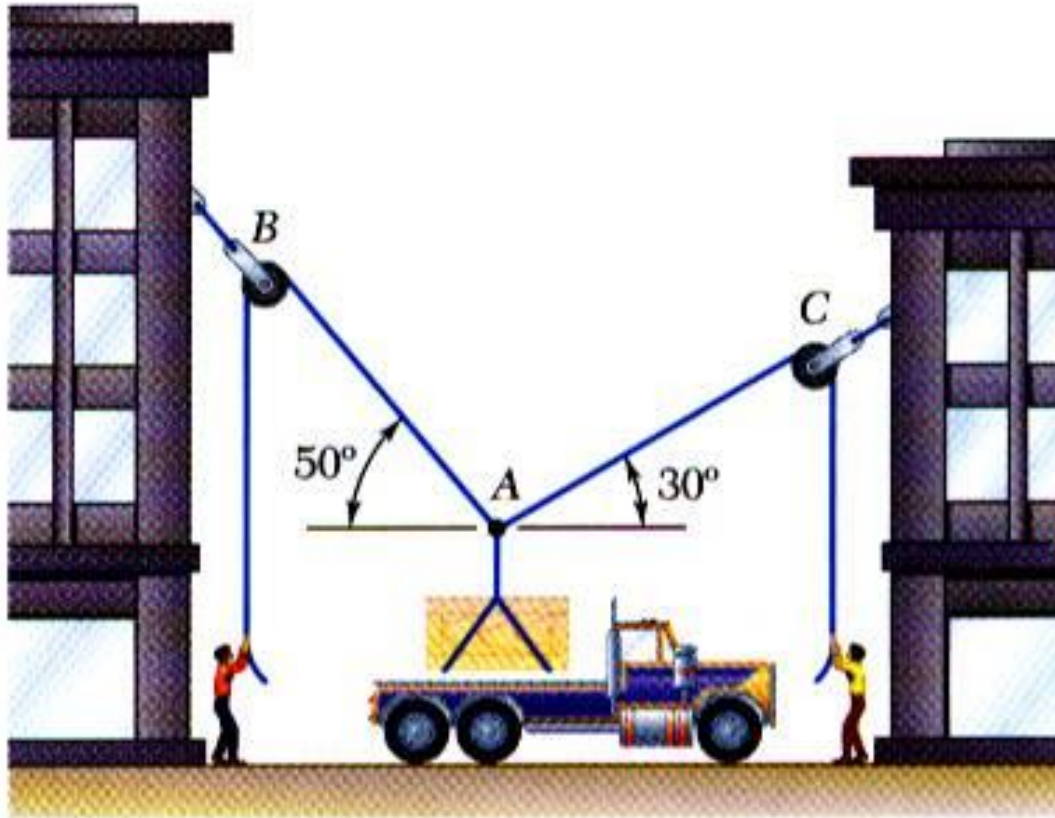
Free-Body Diagram

Free-body Diagram (Further Examples)

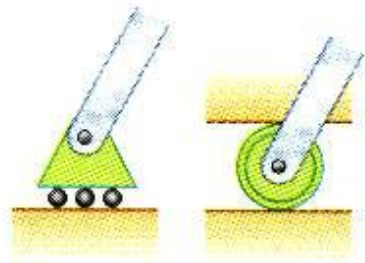
- Disc Suspended by Two Cables



Free-body Diagram (Further Examples)



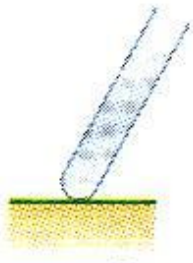
Reactions at Supports and Connections for a 2D Structure



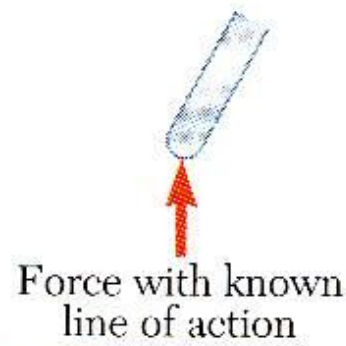
Rollers



Rocker



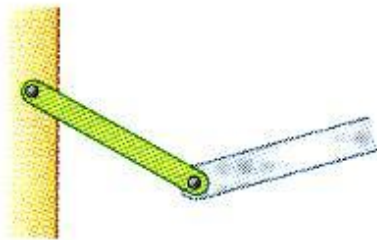
Frictionless surface



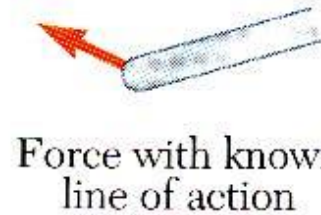
Force with known line of action



Short cable

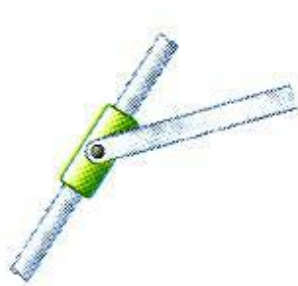


Short link

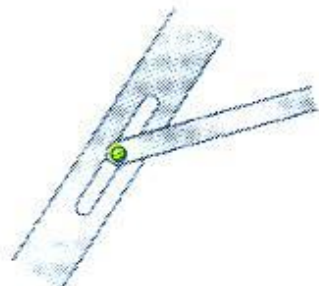


Force with known line of action

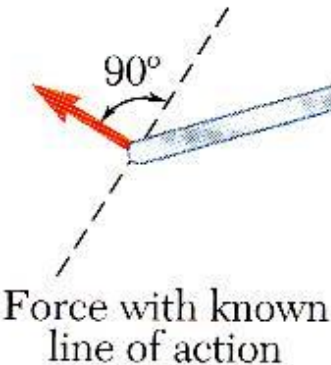
- Reactions equivalent to a force with known line of action.



Collar on frictionless rod



Frictionless pin in slot



Force with known line of action

Coplanar (2D) Force Systems

- If a particle subjected to a system of concurrent coplanar forces is in equilibrium and lie in the x - y plane, then the following conditions must apply:

$$\sum F_x = 0 \quad \text{AND} \quad \sum F_y = 0$$

OR

$$F_{Rx} = 0 \quad \text{AND} \quad F_{Ry} = 0$$

Procedure for Analysis

- **Free Body Diagram**

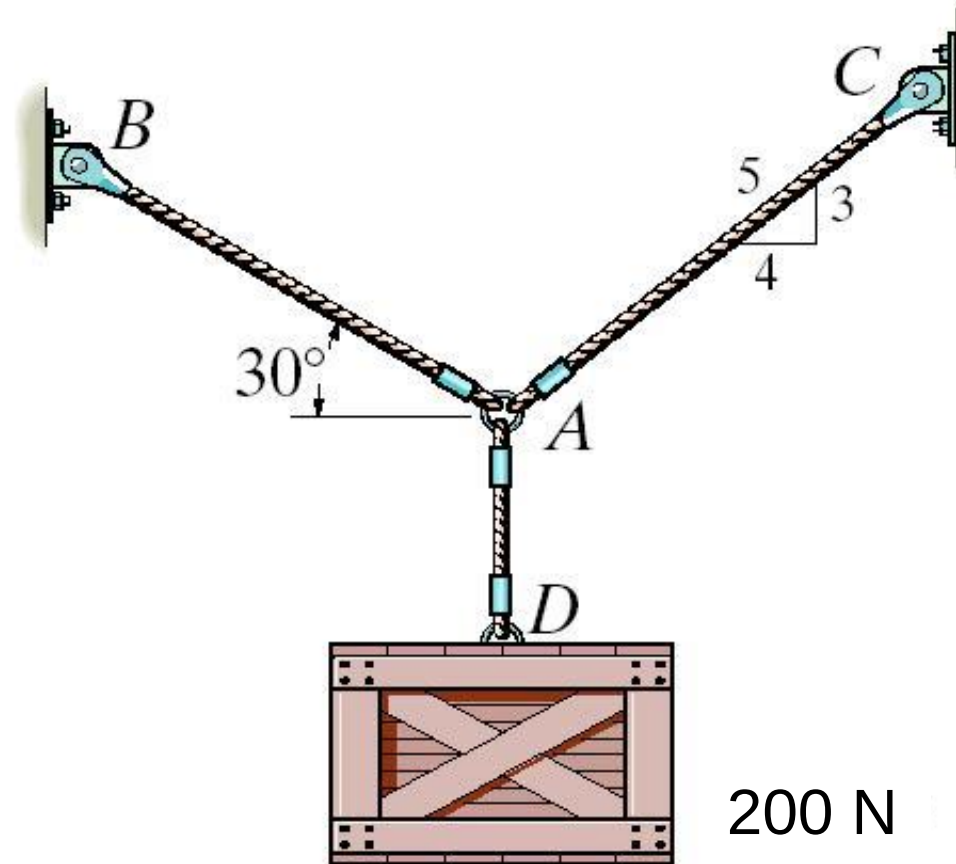
- Establish x , y axes in any suitable orientation
- Label all known and unknown forces and indicate directions
- The sense of unknown forces may be assumed (to be verified at later).

- **Equilibrium Equations**

- Apply equilibrium equations to each Free Body Diagram
- Start with the one with least number of unknowns.

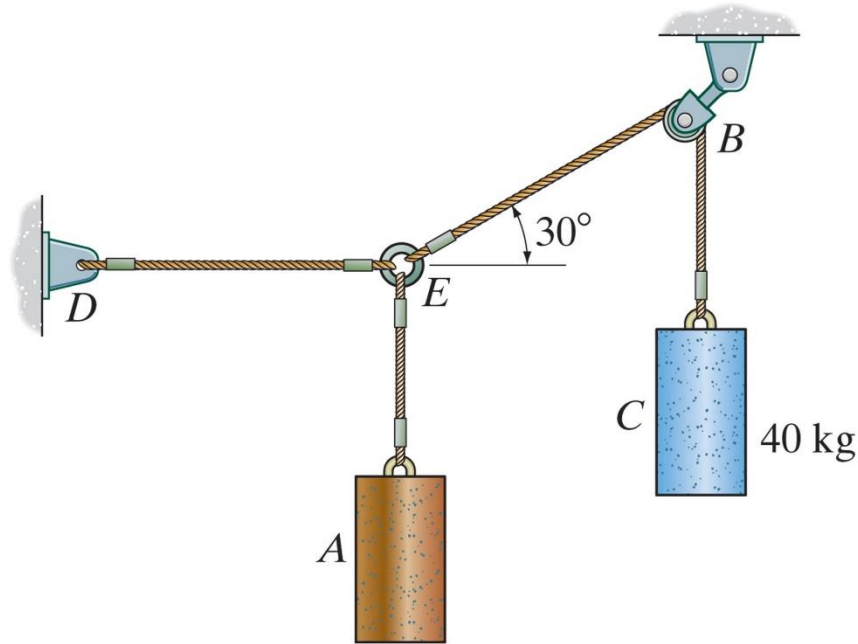
Example 1- Particle Equilibrium (2D)

The box in the figure weighs 200 N. Draw an FBD for point A and the box. Find the tensions in the cables at static equilibrium.



Example 2 –Particle Equilibrium

Determine the mass of bucket A to hold the system in equilibrium if the pulley at B is free to rotate.

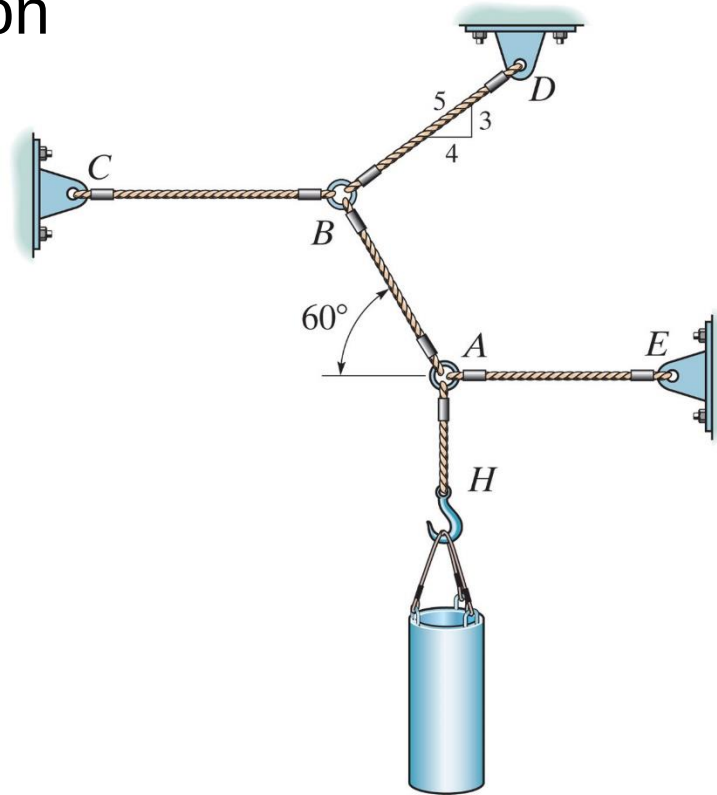


03_FUND-PROB_05

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Example 3 – System of Particles

A pipe of mass 30 kg is held in equilibrium by the system shown. Determine the tension in each rope.



03_PROB_26-27

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Non-coplanar (3D) Force Systems

$$\sum \mathbf{F} = \mathbf{0}$$

OR

$$\mathbf{F}_R = \mathbf{0}$$

OR

$$\sum F_x = 0$$

$$\sum F_y = 0$$

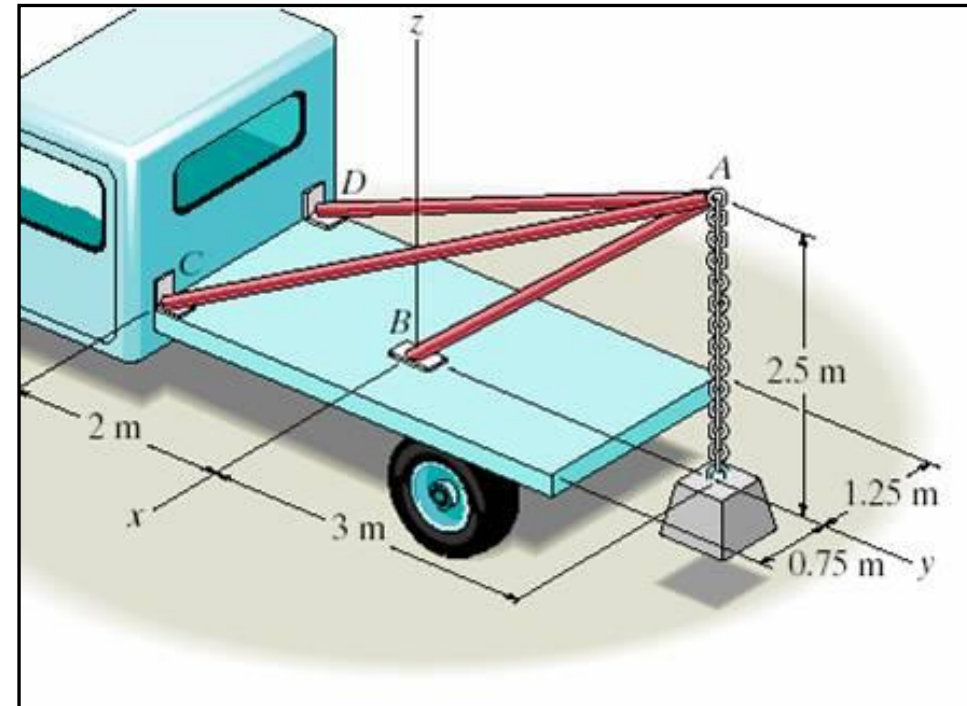
$$\sum F_z = 0$$

OR

$$(F_R)_x = 0$$

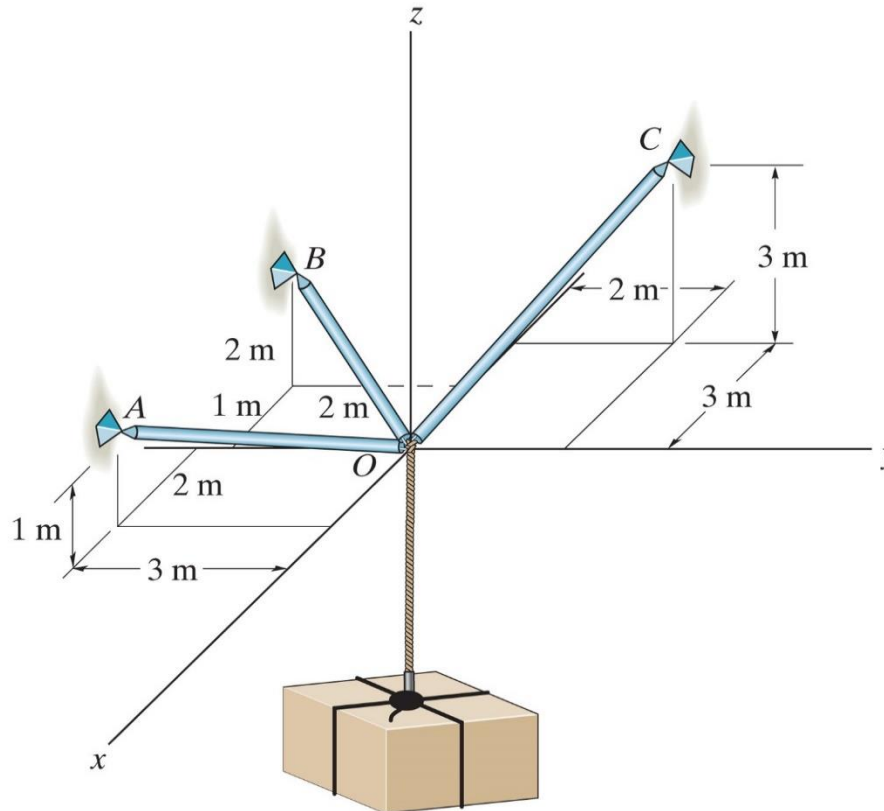
$$(F_R)_y = 0$$

$$(F_R)_z = 0$$



Example 4 – 3D Equilibrium

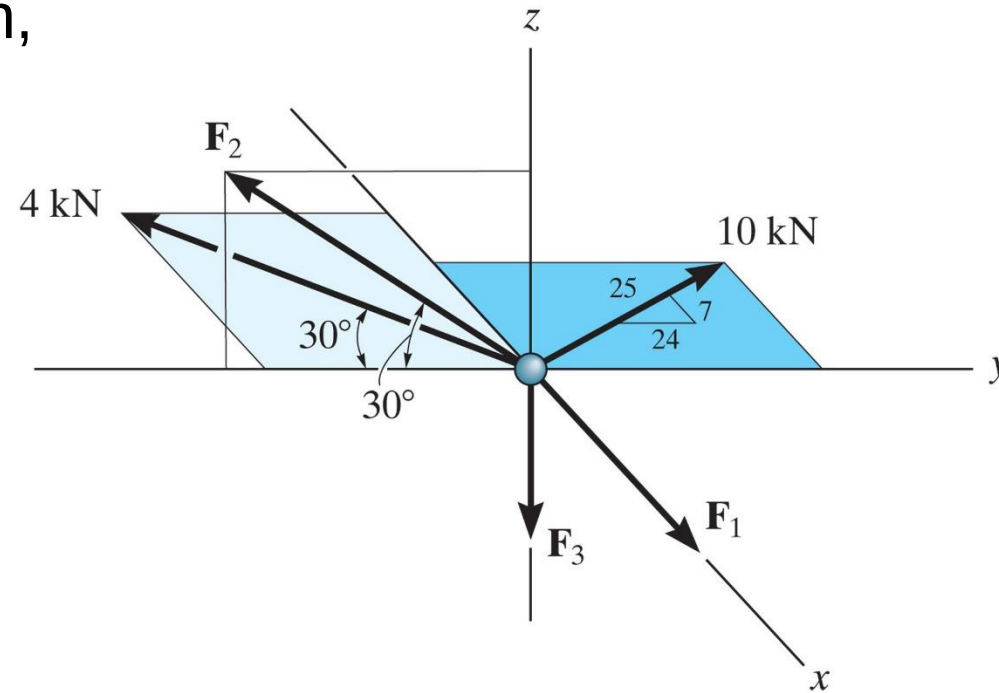
A crate is being held in equilibrium by the system of rods shown here. The maximum force in each rod cannot exceed 1500 N. Determine the greatest mass of the crate that can be supported.



03_PROB_62

Example 5 – 3D Equilibrium

For the system shown,
Find the magnitudes
of F_1 , F_2 and
 F_3 for equilibrium

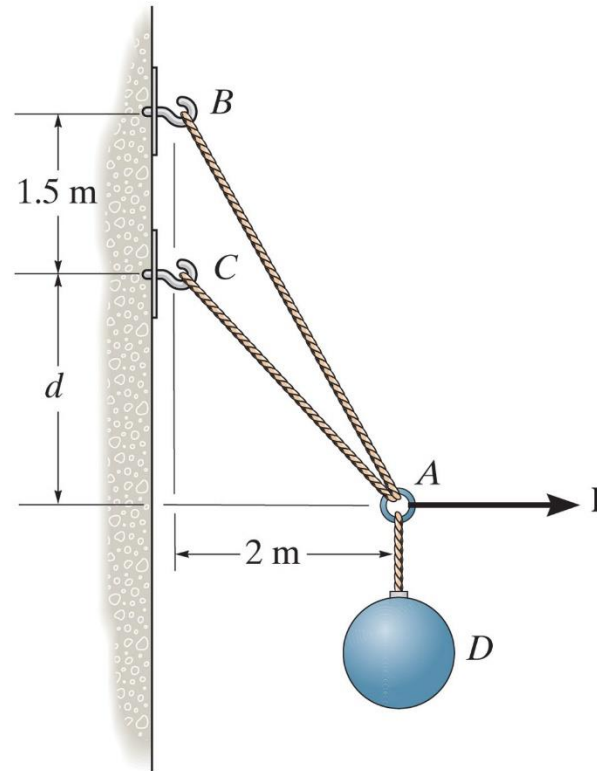


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Assignment 2 – Problem 1

A force of $F = 300\text{ N}$ is needed to keep the 20-kg ball in equilibrium. Determine the Force in cables AB and AC if $d = 1\text{ m}$.

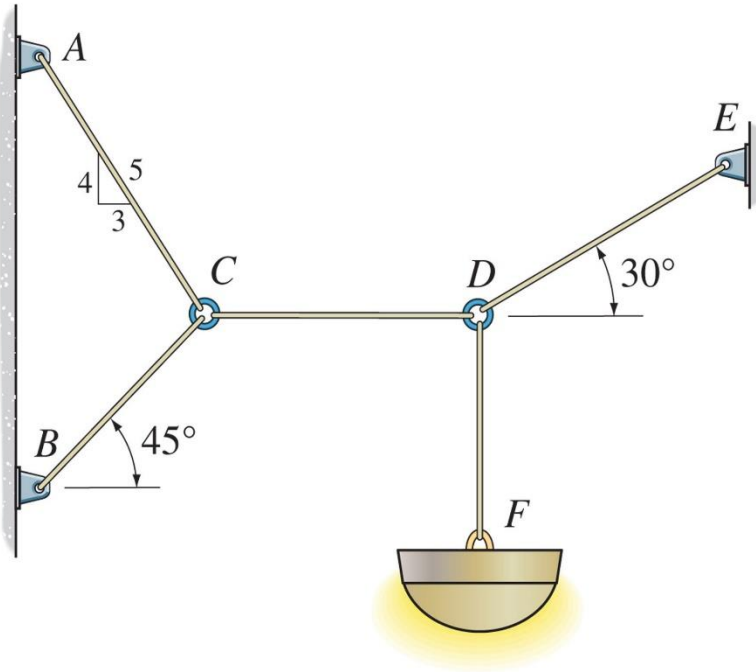


03_PROB_38-39

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Assignment 2-Problem 2

Determine the tension developed in each cord required for equilibrium of the 20 kg lamp

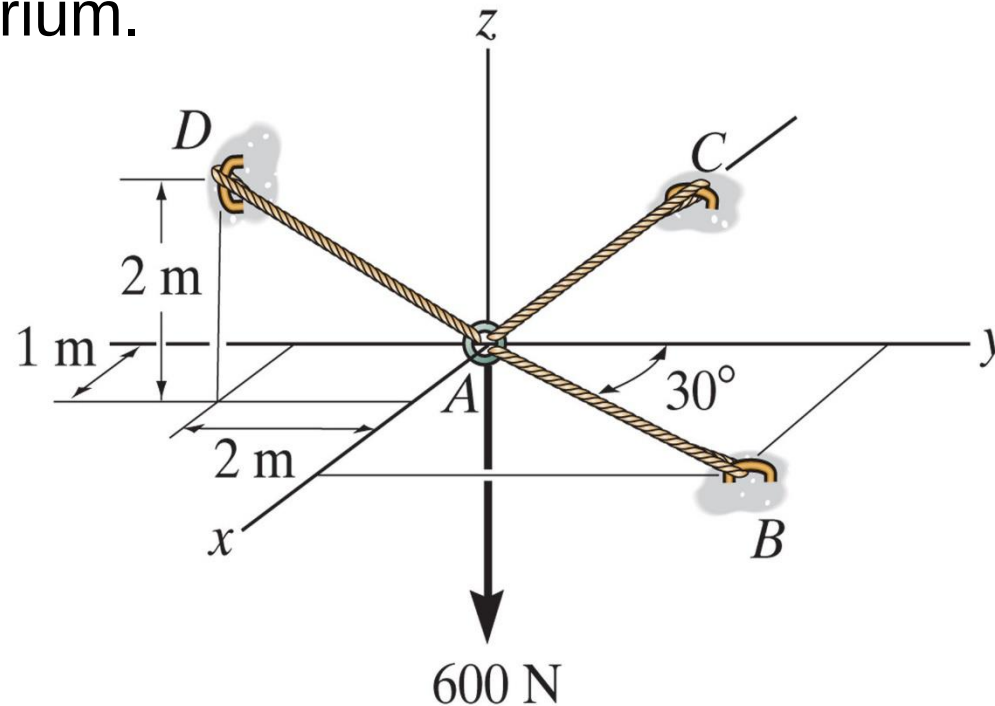


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Assignment 2 – Problem 3

Determine the tension developed in cables AB, AC, and AD at equilibrium.

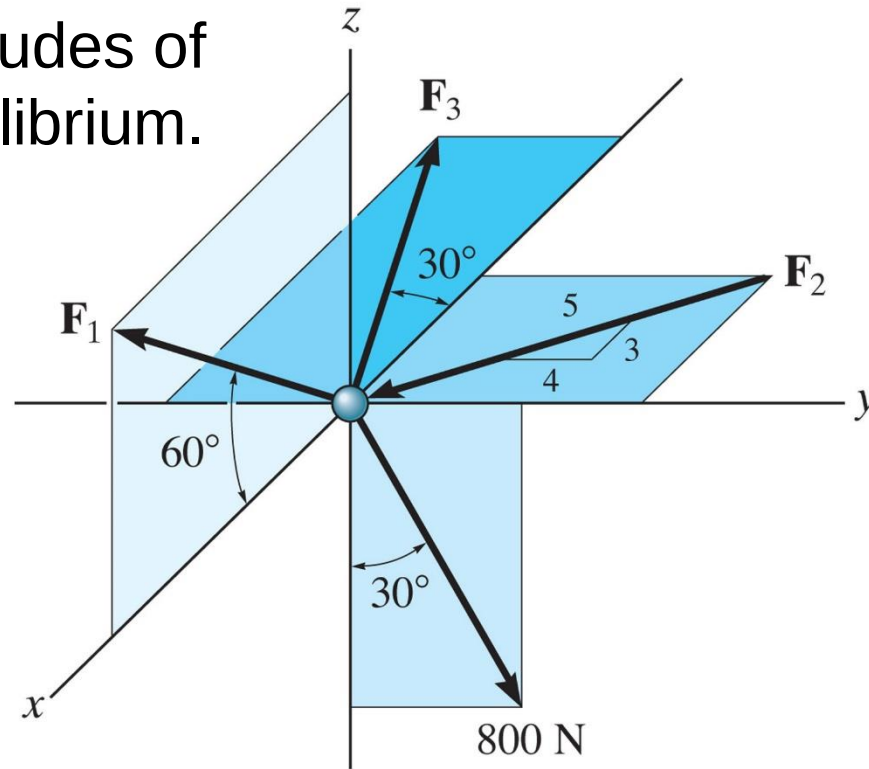


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Assignment 2 – Problem 4

Determine the magnitudes of F_1 , F_2 , and F_3 for equilibrium.

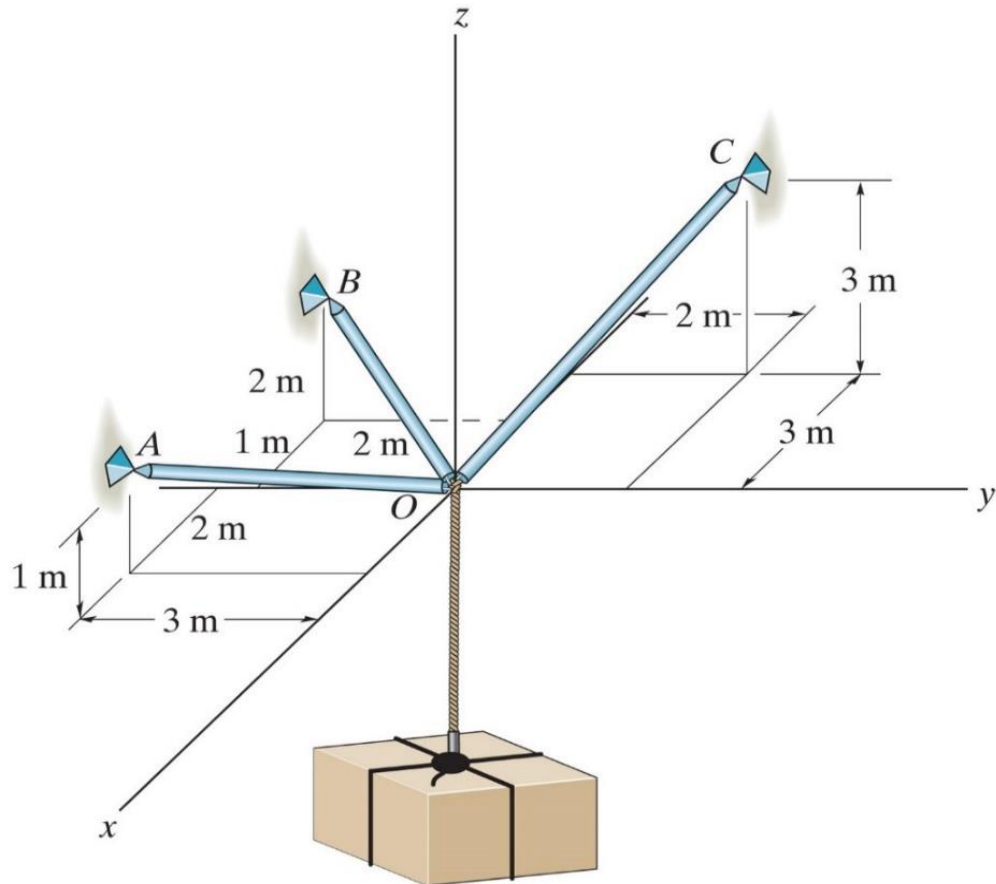


03_REVIEW-PROB_06

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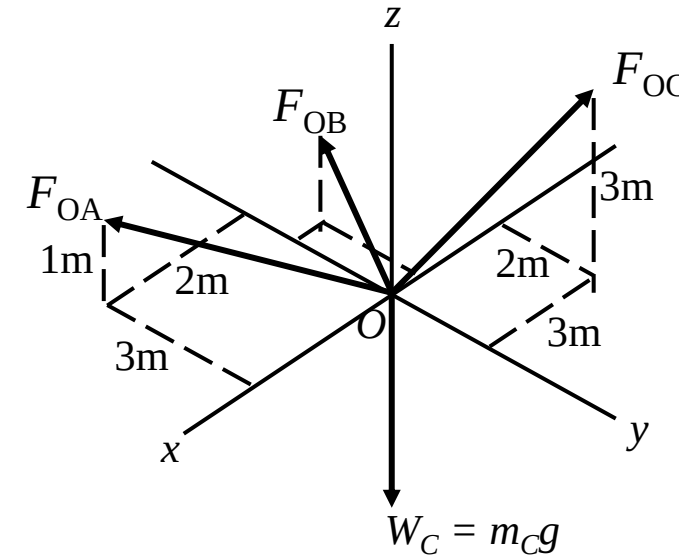
Solved Examples in 3D

- Determine the greatest mass of crate that can be supported if the maximum force in each rod cannot exceed 1500 N.



Solution

Let W_C and m_C denote the mass and weight of the crate, respectively



$$\mathbf{F}_{OA} = F_{OA} \mathbf{u}_{OA}$$

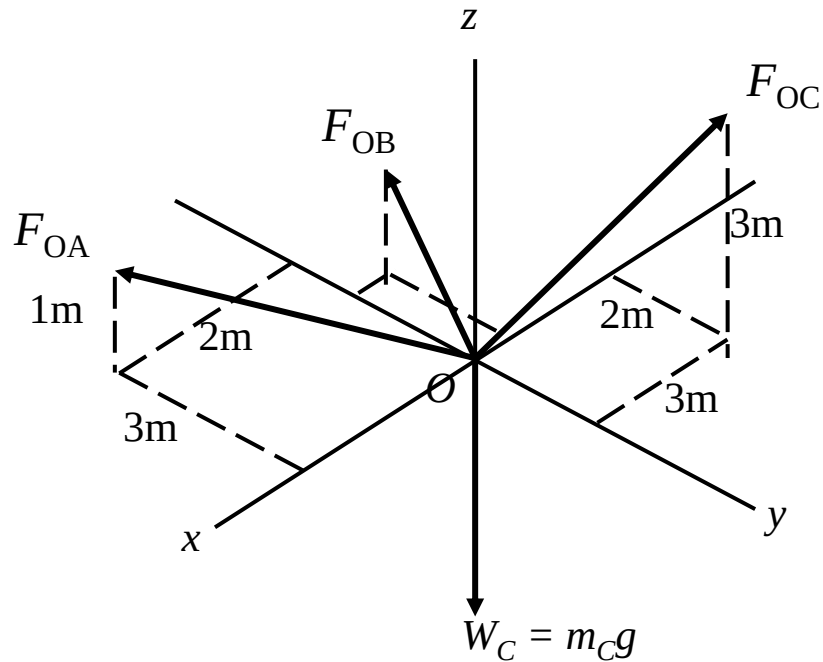
$$\mathbf{F}_{OB} = F_{OB} \mathbf{u}_{OB}$$

$$\mathbf{F}_{OC} = F_{OC} \mathbf{u}_{OC}$$

$$\mathbf{W}_C = -m_C g \mathbf{k}$$

Tutorial Problems 3A, Question #3

Solution



$$\mathbf{u}_{OA} = \frac{\overrightarrow{OA}}{|\overrightarrow{OA}|} = \frac{2\mathbf{i} - 3\mathbf{j} + \mathbf{k}}{\sqrt{[2^2 + (-3)^2 + 1^2]}}$$

$$\mathbf{u}_{OA} = 0.535\mathbf{i} - 0.802\mathbf{j} + 0.267\mathbf{k}$$

$$\mathbf{u}_{OB} = \frac{\overrightarrow{OB}}{|\overrightarrow{OB}|} = \frac{-\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}}{\sqrt{[2^2 + (-3)^2 + 1^2]}}$$

$$\mathbf{u}_{OB} = -0.333\mathbf{i} - 0.667\mathbf{j} + 0.667\mathbf{k}$$

$$\mathbf{u}_{OC} = \frac{\overrightarrow{OC}}{|\overrightarrow{OC}|} = \frac{-3\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}}{\sqrt{[(-3)^2 + 2^2 + 3^2]}}$$

$$\mathbf{u}_{OC} = -0.64\mathbf{i} + 0.426\mathbf{j} + 0.64\mathbf{k}$$

Tutorial Problems 3A, Question #3

At equilibrium, $\sum \mathbf{F} = \mathbf{0}$

$$\begin{aligned} \therefore F_{OA}(0.535\mathbf{i} - 0.802\mathbf{j} + 0.267\mathbf{k}) + F_{OB}(-0.333\mathbf{i} - 0.667\mathbf{j} + 0.667\mathbf{k}) \\ + F_{OC}(-0.64\mathbf{i} + 0.426\mathbf{j} + 0.64\mathbf{k}) - m_C g \mathbf{k} = 0 \end{aligned}$$

$$\begin{aligned} (0.535F_{OA} - 0.333F_{OB} - 0.64F_{OC})\mathbf{i} + (-0.802F_{OA} - 0.667F_{OB} + 0.426F_{OC})\mathbf{j} \\ + (0.267F_{OA} + 0.667F_{OB} + 0.64F_{OC} - 9.81m_C)\mathbf{k} = 0 \end{aligned}$$

$$\Rightarrow 0.535F_{OA} - 0.333F_{OB} - 0.64F_{OC} = 0 \quad (1)$$

$$-0.802F_{OA} - 0.667F_{OB} + 0.426F_{OC} = 0 \quad (2)$$

$$0.267F_{OA} + 0.667F_{OB} + 0.64F_{OC} - 9.81m_C = 0 \quad (3)$$

Solving equations (1), (2) and (3) for F_{OA} , F_{OB} , and F_{OC} in terms of m_C yields

Tutorial Problems 3A, Question #3

$$F_{OA} = 15.463m_C$$

$$F_{OB} = -7.759m_C$$

$$F_{OC} = 16.963m_C$$

What does the
negative sign
mean?



The results indicate that the link with the greatest force is link OC, i.e., $|\mathbf{F}_{OC}| > |\mathbf{F}_{OA}| > |\mathbf{F}_{OB}|$

Hence, link OC will reach the limiting force first

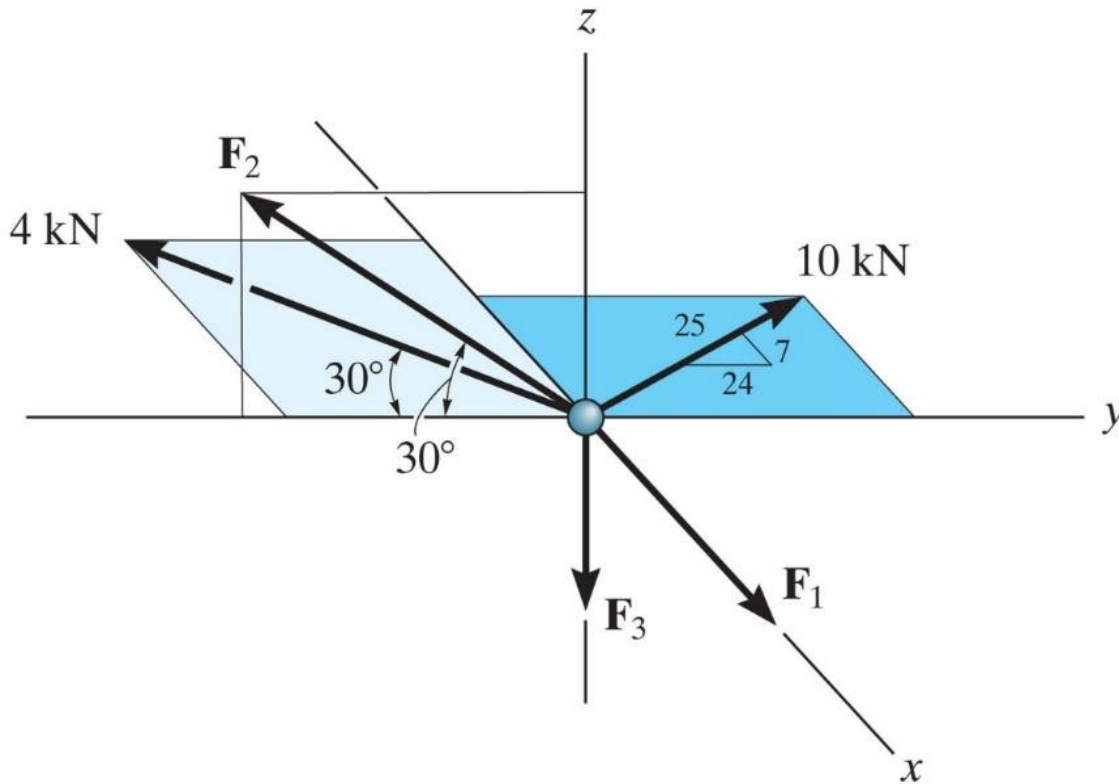
$$\Rightarrow 16.963m_C = 1500$$

$$\therefore m_C = \frac{1500}{16.963} = 88.43\text{kg}$$

Tutorial Problems 3A, Question #3

Solution

- For the system shown, find the magnitudes of F_1 , F_2 and F_3 for equilibrium.



For convenience, let's use the scalar approach

At equilibrium

$$\sum F_x = 0$$

$$\Rightarrow F_1 - 4\sin 30^\circ - 10\left(\frac{7}{25}\right) = 0$$

$$\therefore F_1 = 4.5 \text{ N}$$

$$\sum F_y = 0$$

$$-F_2\cos 30 - 4\cos 30^\circ + 10\left(\frac{24}{25}\right) = 0$$

$$\therefore F_2 = 7.085 \text{ N}$$

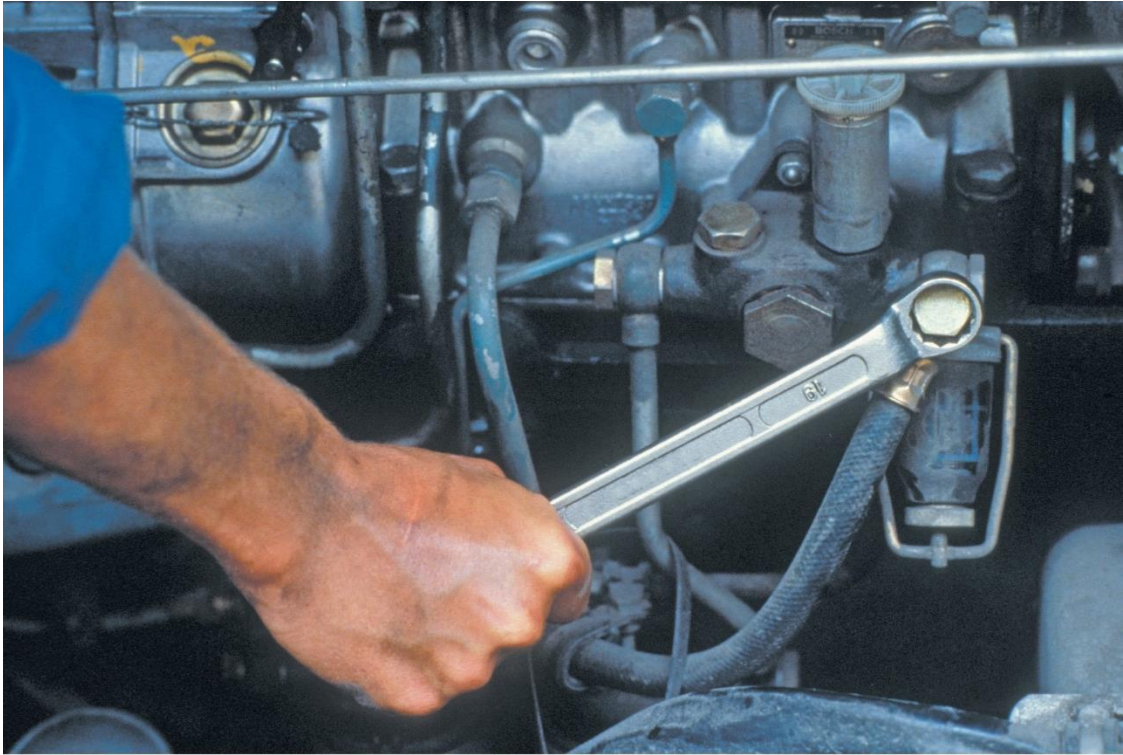
Moments of a Force



Contents

- Moment of a Force: Scalar Formulation
- Vector III
- Moment of a Force: Vector Formulation
- Moment of a Force about a Specified Axis
- Moment of a Couple
- Simplification of a Force and Couple System
- Reduction of a Simple Distributed Loading

Applications



The force applied to the wrench will produce rotation or a tendency for rotation



The ability to remove the nail will depend on the moments F_H and F_N about O produces

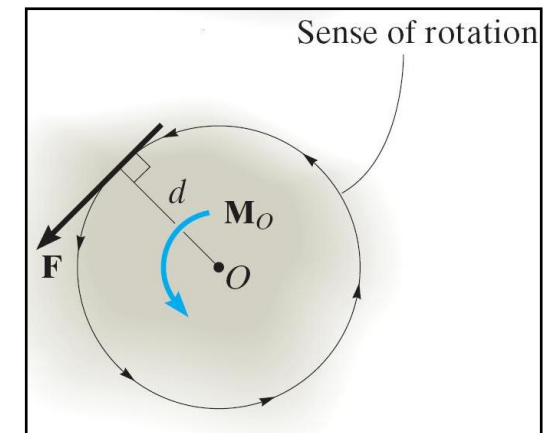
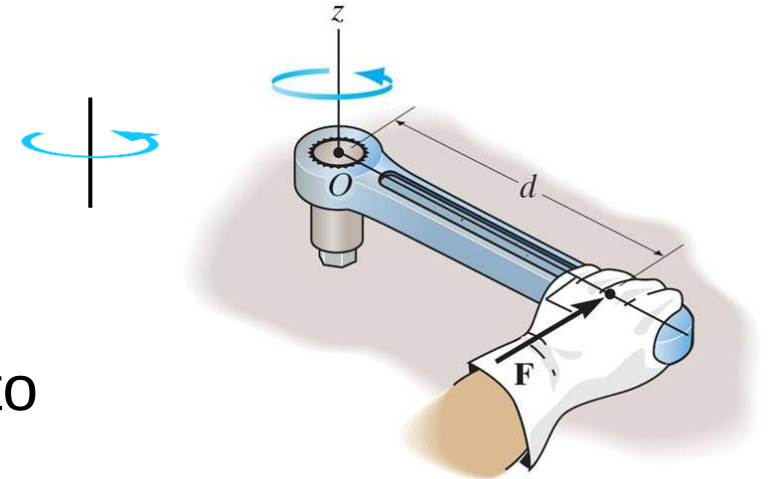
Moment of a Force: Scalar Formulation

- The moment of a force (moment or torque) about a point or about an axis provides a measure of the tendency for rotation.
- The magnitude, M_o , is calculated as:

$$M_o = Fd$$

d = *moment arm* or perpendicular distance from the Point to the Line of action of the force

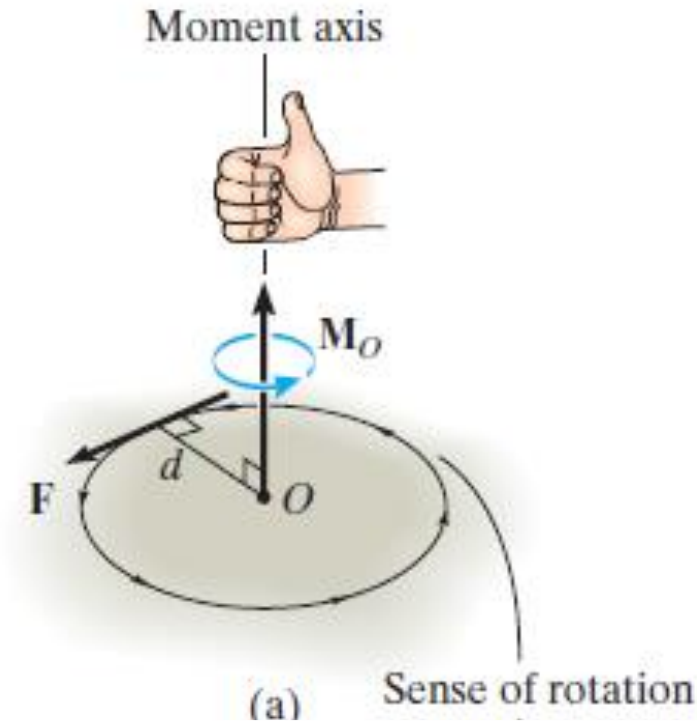
- The direction of $\mathbf{M}_o$ is defined by its moment axis.
 - It is perpendicular to the plane of the force and the moment arm
 - The right-hand rule is used to determine the direction of $\mathbf{M}_o$



Important Results

A Moment has both Magnitude and Direction:

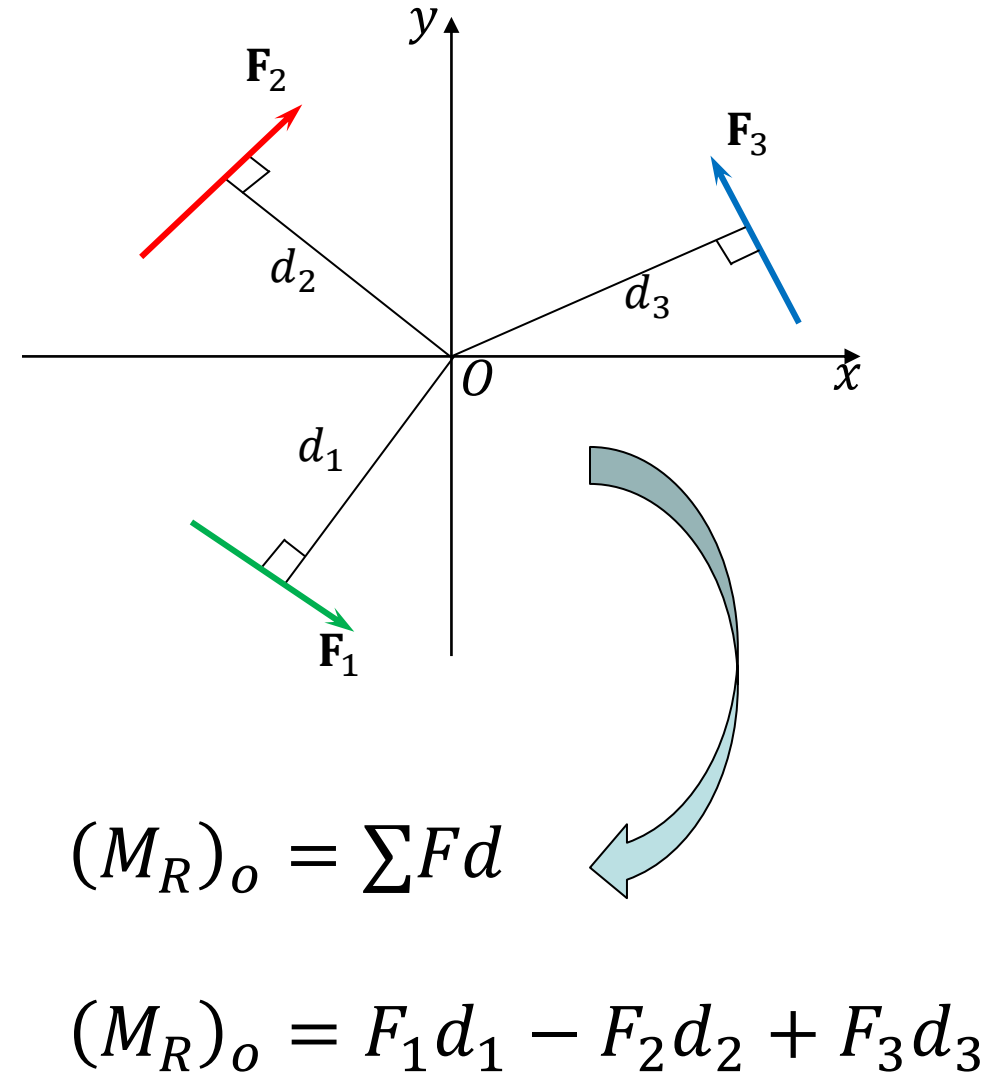
- The direction is determined using the right-hand rule. That is, if the curl of the fingers of the right hand represent direction of rotation of the moment (clockwise or counterclockwise), then the right thumb gives the direction of the moment.
- Thus, a moment can be treated as a vector quantity
- Counterclockwise moments are positive because they point in the direction of the positive z axis
- Clockwise moments are negative because they point in the direction of the negative z axis



Important Results

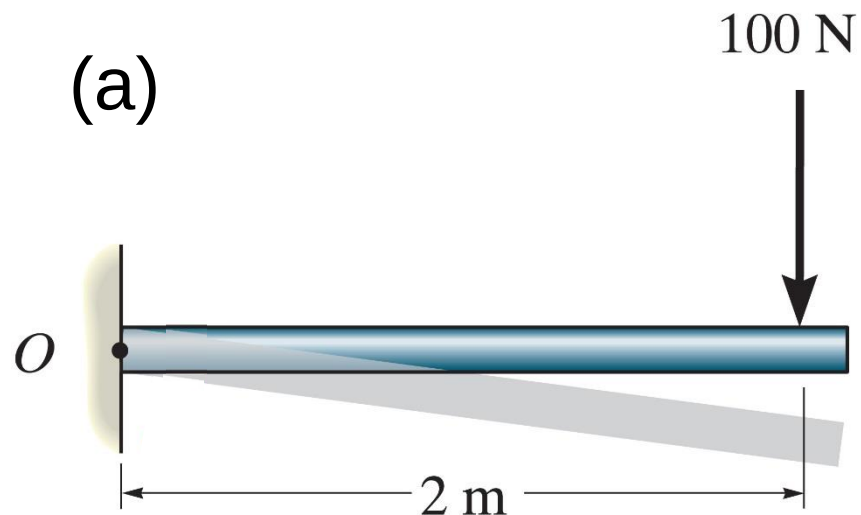
Resultant moment:

- For 2D problems where all the forces lie in the same plane, the scalar formulation is more convenient to use.
- The resultant moment $(M_R)_O$ about the point O is formulated as the algebraic sum of all moments about O
 - Clockwise moments are -ve
 - Counterclockwise moments are +ve

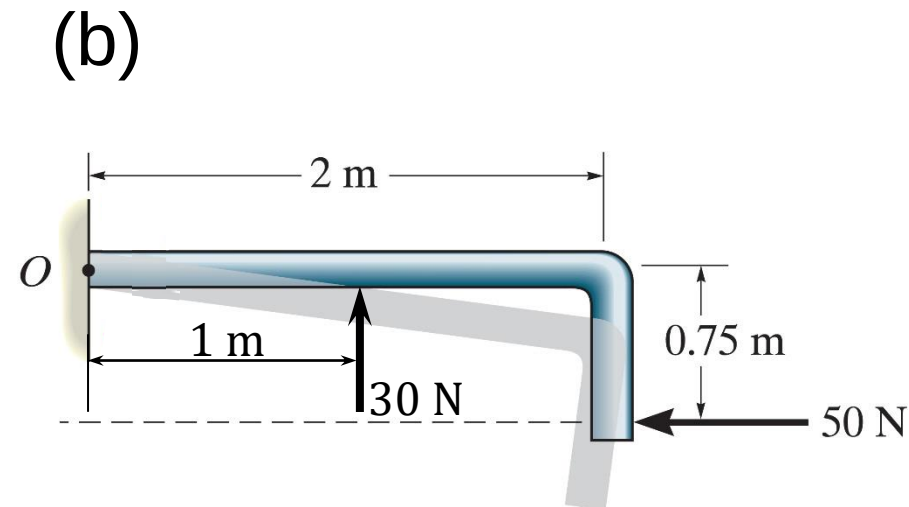


Example 1 (a, b)

Find Moment about O for each



$$\begin{aligned}M_o &= -(100 \text{ N}) \times (2 \text{ m}) \\&= -200 \text{ Nm}\end{aligned}$$



$$\begin{aligned}M_o &= \sum Fd \\ \therefore M_o &= (30 \times 1) - (50 \times 0.75) \\ &= -7.5 \text{ Nm}\end{aligned}$$

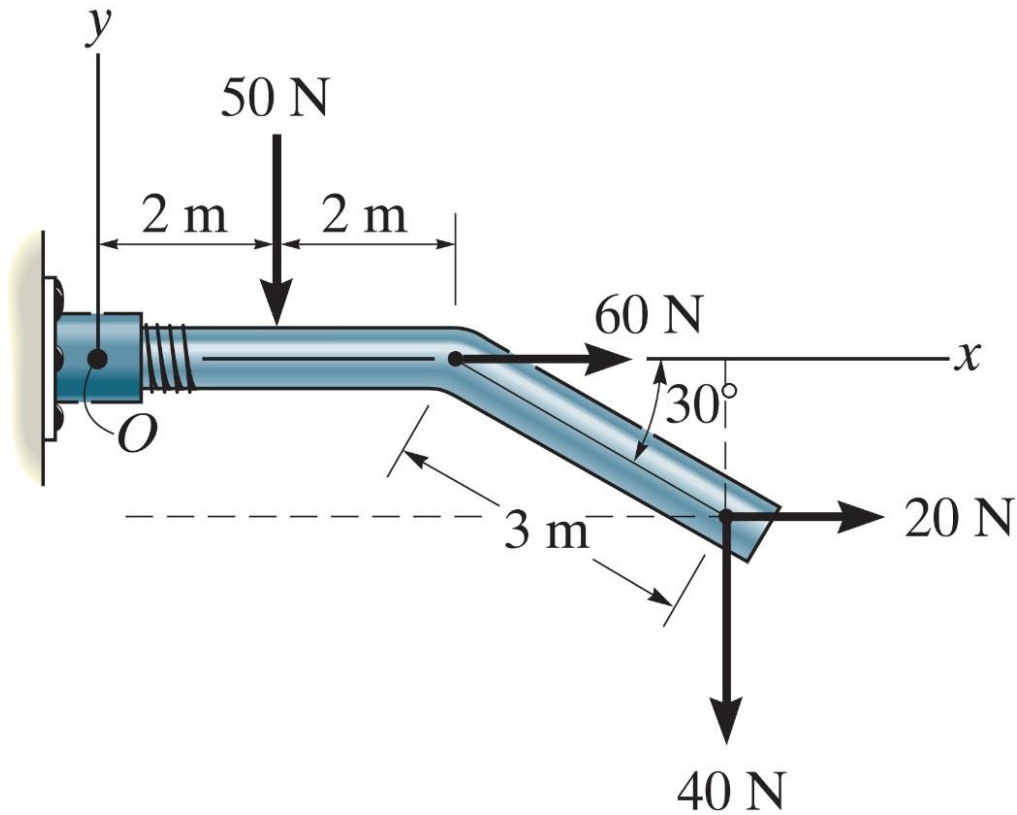
Example 1c

Determine the resultant moment of the force(s) about point O.

Solution

$$(M_R)_O = \sum Fd$$

$$\begin{aligned}(M_R)_O &= -(50 \times 2) + (60 \times 0) \\ &\quad - [40 \times (4 + 3\cos 30^\circ)] \\ &\quad + 20 \times 3\sin 30^\circ \\ &= -333.92 \text{ Nm}\end{aligned}$$



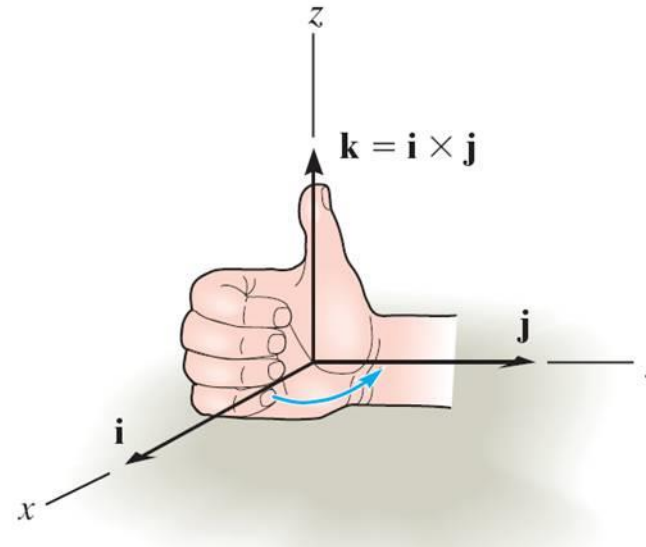
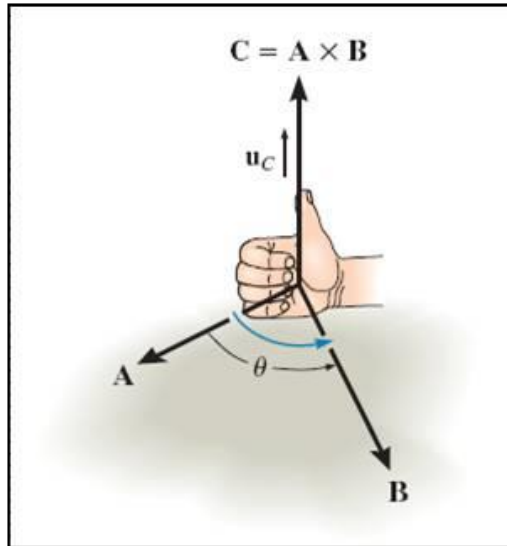
Moment of a Force: Vector Formulation

■ Cross Product

- The cross product of two vectors **A** and **B** results in another vector, **C**, given as:

$$\mathbf{C} = \mathbf{A} \times \mathbf{B} = AB\sin\theta\mathbf{u}_C$$

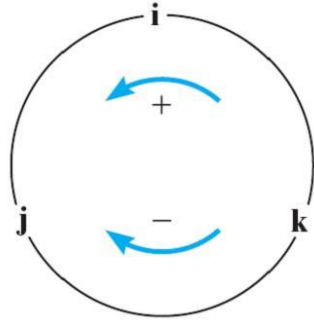
- The direction of **C** is perpendicular to the plane containing **A** and **B**, which can be specified by the right-hand rule.



Example 2 – Cross Product

- Cross Product

The cross product operations can be carried out “directly” or by writing in a compact determinant form and expanding, as:



$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Determine $\mathbf{A} \times \mathbf{B}$ using vector algebra and the matrix form.

(a) $\mathbf{A} = -5\mathbf{i}$ and $\mathbf{B} = 6\mathbf{i} + 4\mathbf{j} - 10\mathbf{k}$

(b) $\mathbf{A} = 2\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 2\mathbf{k}$

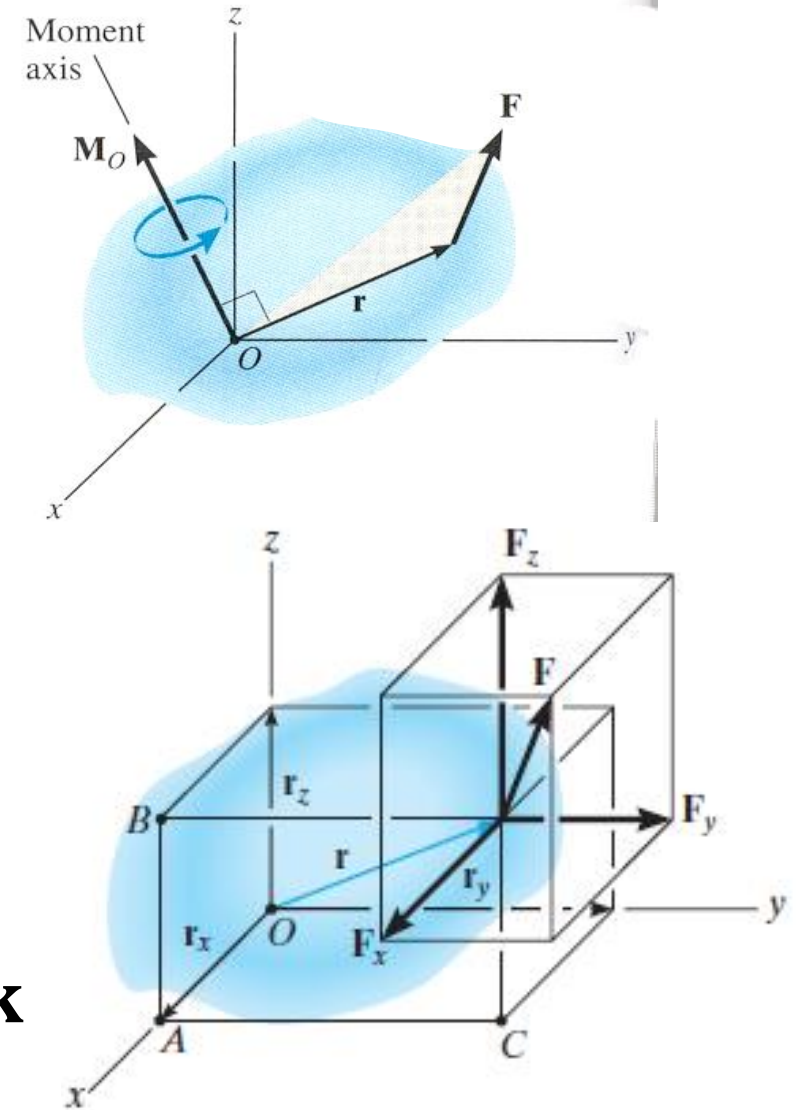
Moment of a Force: Vector Formulation

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$\mathbf{r}$ = position vector directed from O to any point on the line of action of $\mathbf{F}$.

- If r_x , r_y , and r_z and F_x , F_y and F_z represent the Cartesian components of the position vector $\mathbf{r}$ and force vector $\mathbf{F}$, respectively, then

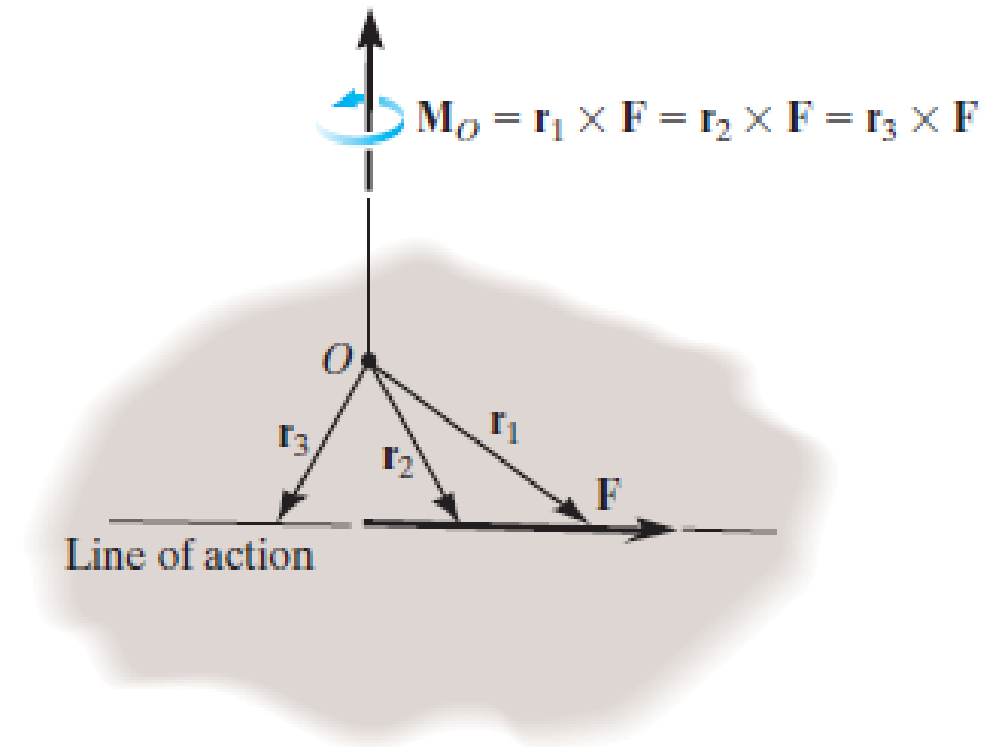
$$\mathbf{M}_O = (r_y F_z - r_z F_y)\mathbf{i} - (r_x F_z - r_z F_x)\mathbf{j} + (r_x F_y - r_y F_x)\mathbf{k}$$



Moment of a Force: Vector Formulation

- Note that in 3D the perpendicular distance of the force vector is not needed
- In 3D all that is required is the position vector from the point O to any point on the line of action of the force
- **The Principle of Transmissibility:** It states that a force acting on a body is a sliding vector since it can be applied at any point along its line of action.

$$\mathbf{M}_O = \mathbf{r}_1 \times \mathbf{F} = \mathbf{r}_2 \times \mathbf{F} = \mathbf{r}_3 \times \mathbf{F}$$



Moment of a Force: Vector Formulation

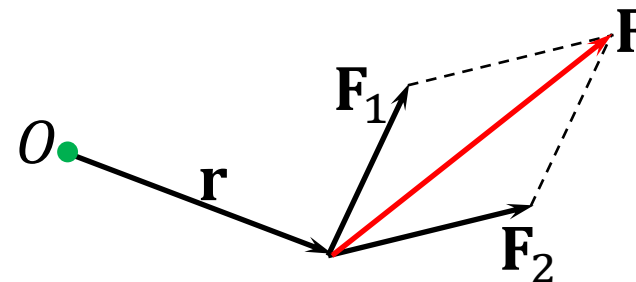
- Resultant moment of a system of forces about point O , is given as:

$$(\mathbf{M}_R)_O = \sum (\mathbf{r} \times \mathbf{F})$$

- **Principle of moments:** The moment of a force about a point is equal to the sum of moments of the components of the force about the point. Also known as Varignon's theorem.

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2$$

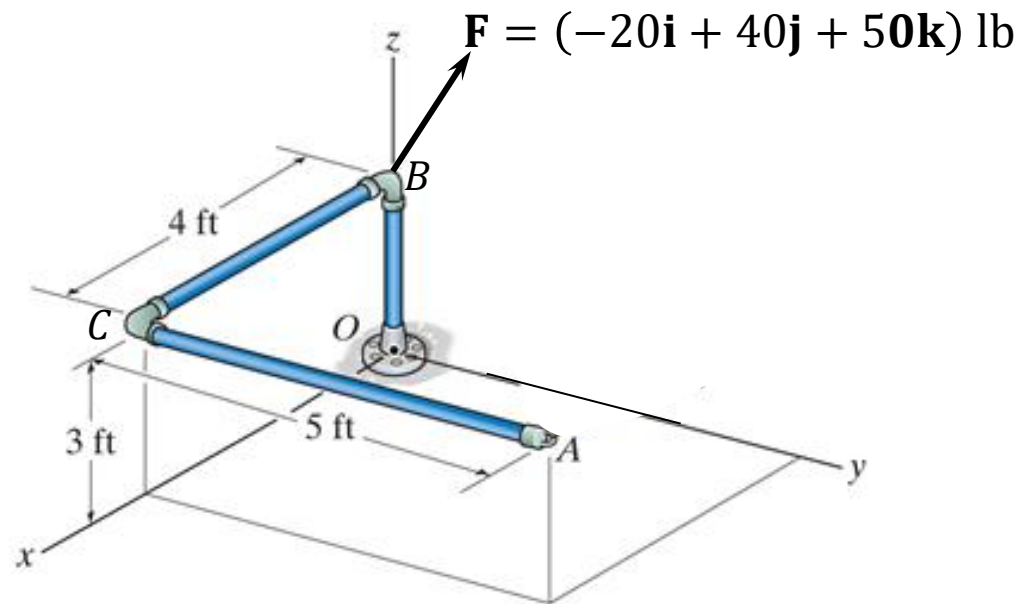
$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2)$$



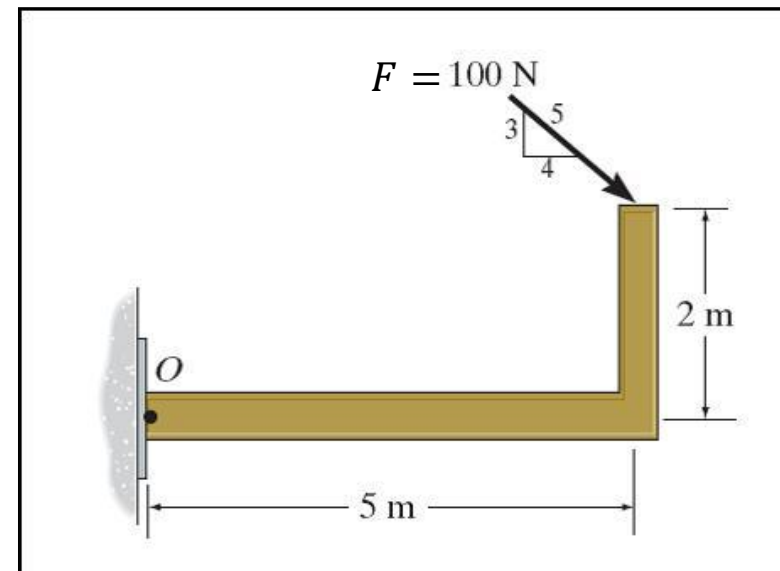
Example 3

Using vector approach, determine the moment produced by the force $\mathbf{F}$ in the figure about O .

(a)

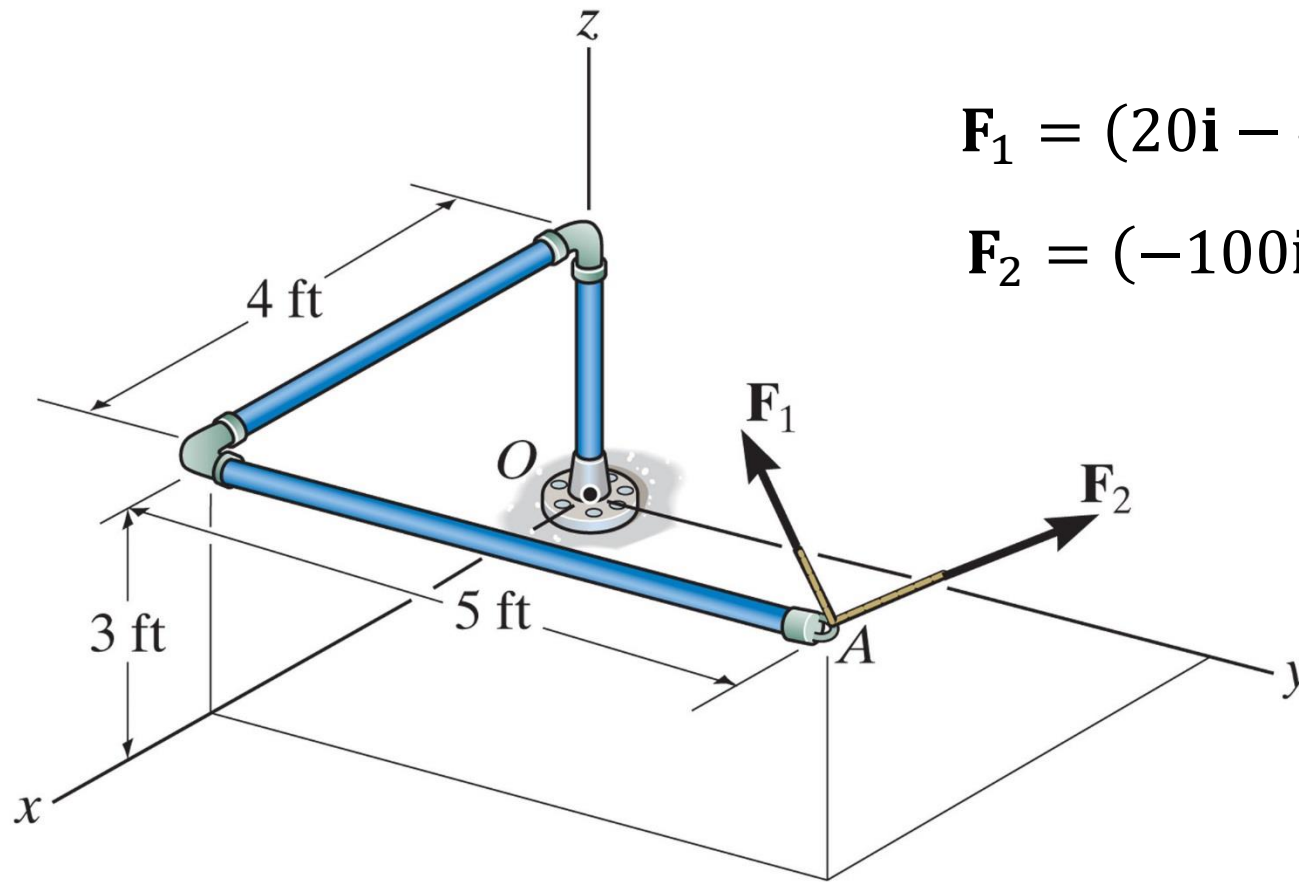


(b)



Example 4

Determine the magnitude of the resultant moment of the forces in the figure about O .



$$\mathbf{F}_1 = (20\mathbf{i} - 40\mathbf{j} + 50\mathbf{k}) \text{ lb}$$

$$\mathbf{F}_2 = (-100\mathbf{i} + 10\mathbf{j} + 50\mathbf{k}) \text{ lb}$$